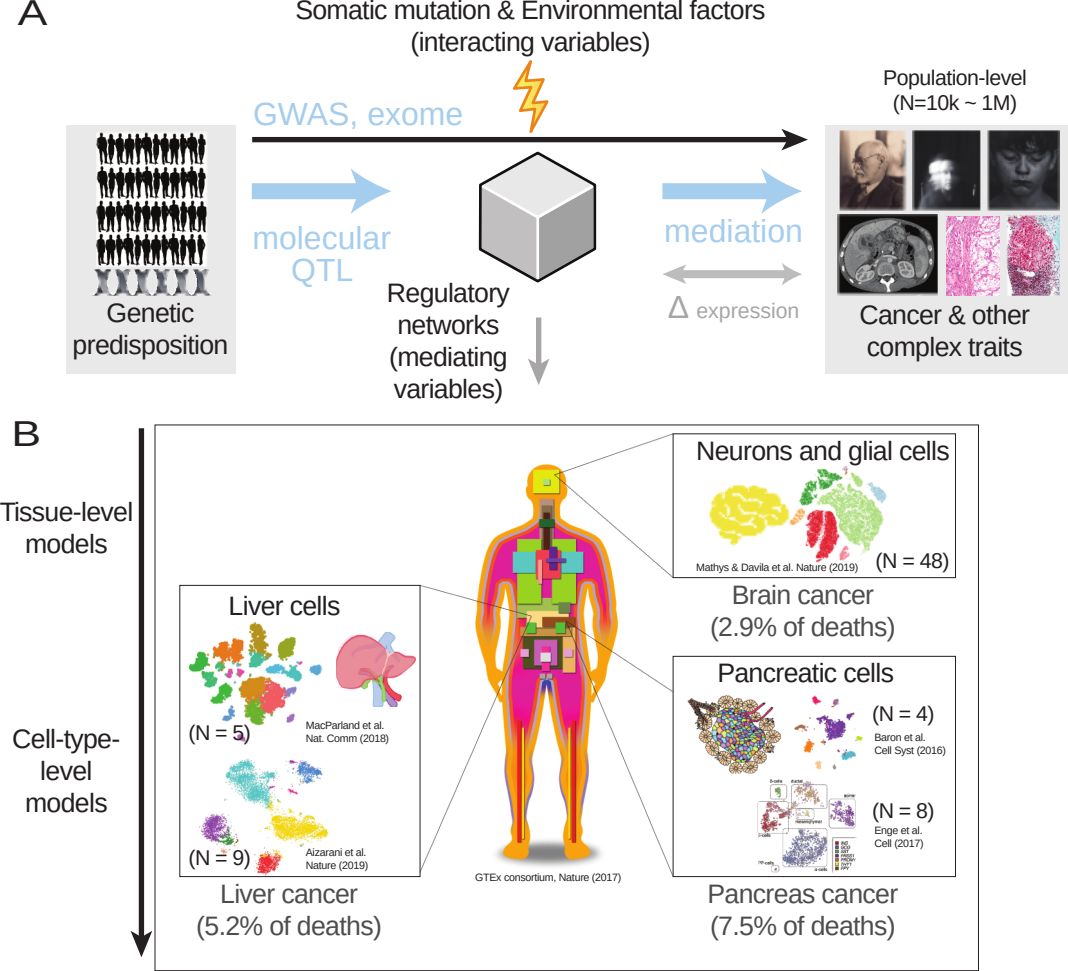
A
Somatic mutation & Environmental factors
(interacting variables)
GWAS, exome
Population-level
(N=10k ~ 1M)
Genetic
predisposition
molecular
QTL
Regulatory
networks
(mediating
variables)
mediation
Δ expression
Cancer & other
complex traits
B
Tissue-level
models
Cell-type-
level
models
Neurons and glial cells
Mathys & Davila et al. Nature (2019)
(N = 48)
Brain cancer
(2.9% of deaths)
Liver cells
(N = 5)
MacParland et al.
Nat. Comm (2018)
(N = 9)
Aizarani et al.
Nature (2019)
Liver cancer
(5.2% of deaths)
GTEx consortium, Nature (2017)
Pancreatic cells
(N = 4)
Baron et al.
Cell Syst (2016)
(N = 8)
Enge et al.
Cell (2017)
Pancreas cancer
(7.5% of deaths)